

SERC Project Workflow & Team Assignments

This document outlines who owns each module, what they are responsible for, and how the team should move through the project without stepping on each other's code.

1. Team Ownership

Phase	Owner	Module	Key Deliverables
0: The Architect	Member 3	serc_types.h	PCB struct, MemBlock struct, global constants
1: The UI / Utility	Member 1	logger.c, ui.c	Logging wrappers, validation, menu helpers
2: The Engine	Member 2	process.c, scheduler.c	FCFS, Round Robin, virtual clock, Gantt charts
3: The Memory	Member 3	memory.c	Best Fit, Worst Fit, fragmentation reports
4: The Features	Member 4	ipc.c, deadlock.c	Banker's Algorithm, shared memory, pipes
5: The Closer	Member 1	main.c	Main menu loop, switch statements, integration

2. Suggested Day-by-Day Plan

Day	Focus	Who Works	Expected Result
Day 1	Lock the foundation	Member 3	Finish serc_types.h and freeze all shared structs/constants
Day 2	Build utilities	Member 1	Complete logger.c and ui.c so the rest of the team can reuse
Day 3	Start the engine	Member 2	Implement process creation and FCFS scheduling
Day 4	Memory integration	Member 2 + Member 3	Connect process creation to memory allocation
Day 5	Advanced scheduling	Member 2	Finish Round Robin and Gantt chart generation
Day 6	Independent features	Member 4	Build IPC demo and Banker's Algorithm
Day 7	Integration day	Member 1 + Everyone	Connect all modules in main.c and fix compile errors
Day 8	Testing and cleanup	Entire Team	Run full project, debug edge cases, polish output

3. Dependency Flow

The order matters. Member 3 must finish the shared header file first. If the PCB struct changes halfway through the sprint, the rest of the project breaks. After that, Member 1's logging and UI code gives everyone reusable tools. Member 2 and Member 3 then work closely because process creation depends on memory allocation. Member 4 can work more independently because IPC and deadlock handling are mostly standalone features. The final stage is integration, where Member 1 pulls everything together in main.c.

4. Important Notes

- Member 1 should create the Makefile so the whole project builds with a single make command.
- Member 2 should use a virtual clock variable instead of real time to make debugging easier.
- Member 4 should use actual Linux IPC system calls like shmget, msgget, pipe, and fork.
- No one should edit serc_types.h after Day 1 unless the whole team agrees on the change.